

THERMOCHEMICAL SIMULATION v0: INTERIM RESULTS FOR THE NEW PHENOLIC LINER

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SEPTEMBER 24, 2026



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Purpose of this document. This report records the reproducible preparation of the first model version for the new motor. It strictly separates the requested geometry from assumptions inherited from the earlier simulations. The native Windows model, mesh, restart chain, and MAPDL preflight are validated. The transient calculation is running; this edition freezes accepted results at 5.2500 s, index 405. The primary goal is to compute, without ambiguous terminology, removed phenolic thickness, remaining thickness, pyrolysis-gas mass, and consumed-carbon mass at every time.

1 Definition of the new motor

1.1 Imposed dimensions

The brief supplied for this study contains three diameters: $D_{\text{Al,out}} = 6.000$ in, $D_{\text{Al,in}} = D_{\text{ph,out}} = 5.625$ in, and $D_{\text{ph,in}} = 5.250$ in. Unlike the first two historical simulations, these dimensions are correctly treated as *diameters* and divided by two before they become radial coordinates.

Table 1. Nominal radial interfaces.

Interface	Diameter [in]	Diameter [mm]	Radius [mm]
Phenolic bore	5.250	133.350	66.6750
Phenolic/aluminium	5.625	142.875	71.4375
Aluminium exterior	6.000	152.400	76.2000

The result is one 4.7625 mm phenolic layer and an aluminium shell of the same thickness. There is no epoxy, second phenolic layer, or artificial intermediate contact. The simulation uses a 50 mm axial coupon and a 0.10° sector, giving a 3,600 scale factor to the full ring of the same length. Because the actual motor length was not supplied, no total motor mass is extrapolated beyond this 50 mm segment.

New nominal geometry – radial cross-section

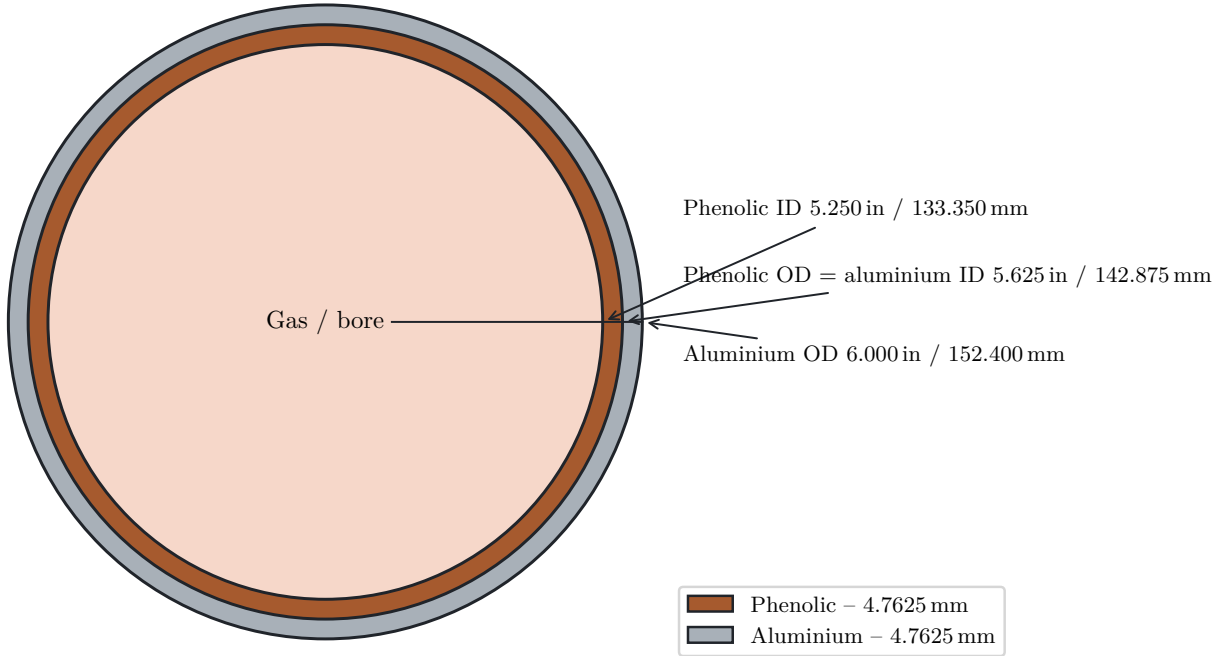


Figure 1. Nominal two-layer cross-section.

As an analytical check, the full 50 mm segment contains $V_{\text{ph},0} = 1.033\,208 \times 10^{-4} \text{ m}^3$, or $m_{\text{ph},0} = 0.129\,151 \text{ kg}$ at the reference density $\rho_v = 1250 \text{ kg m}^{-3}$. The corresponding sector mass is 35.8753 mg. These are balance references, not measurements of the manufactured part.

1.2 Geometry files

The `geometry/newmotor_phenolic_v0.db` database is a native MAPDL 2026 R1 file containing both glued volumes. It was created on Windows without a mesh or a `SOLVE` command. Its standalone, readable generator, `model/generate_geometry_mapdl.inp`, remains the geometry source. The exchange database uses global Z as its axis; the analytical solver mesh uses global Y. A rigid rotation relates them and changes no dimension or thermal condition.

2 Computational domain and mesh

2.1 Discretization choices

The mesh is structured, conformal at the interface, and contains only linear SOLID278 thermal hexahedra. The two materials share interface nodes, so no contact elements are required. The phenolic has 240 radial rows:

$$\Delta r_{\text{ph}} = \frac{4.7625 \text{ mm}}{240} = 19.84375 \mu\text{m}.$$

The aluminium has 64 rows of $74.4141 \mu\text{m}$. The axis is divided into 250 intervals of 0.20 mm and the angle into two intervals. Each potentially exposed phenolic row therefore has 500 faces.

Conformal structured mesh validated in MAPDL

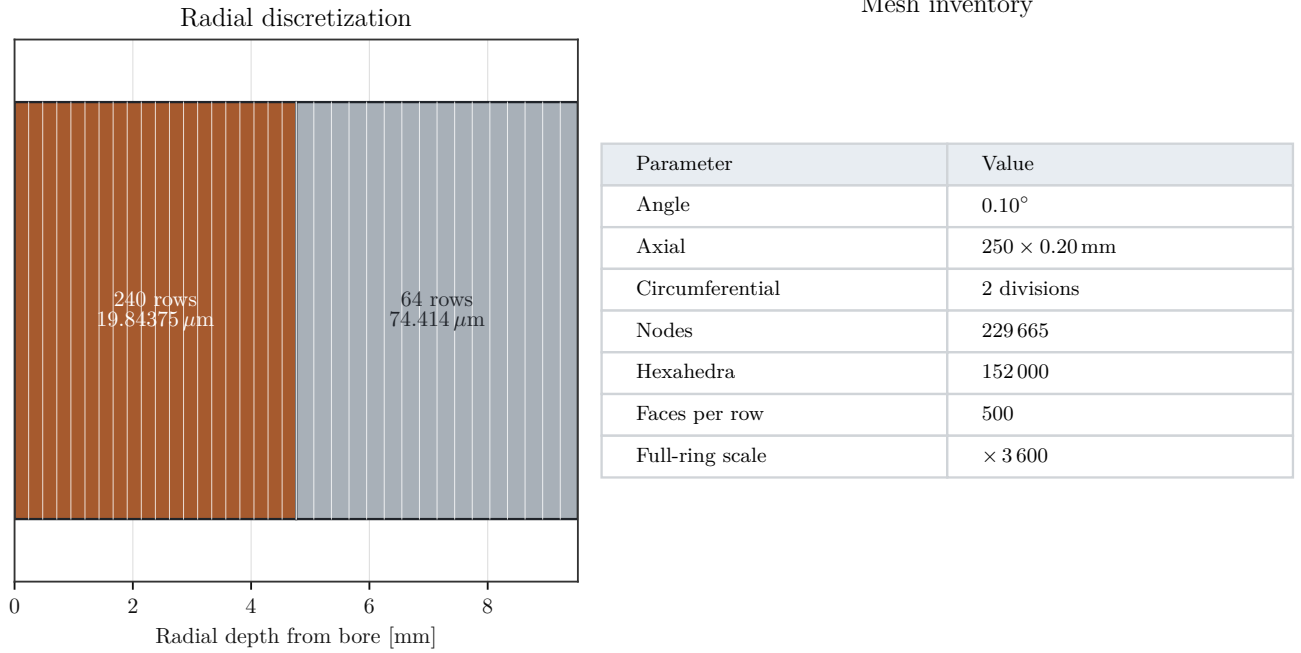


Figure 2. Radial resolution and final mesh inventory.

The final model has 229,665 nodes and 152,000 hexahedra: 120,000 in phenolic and 32,000 in aluminium. The first 100-division axial mesh produced an aspect ratio of 25.2 and was archived in the temporary directory. Increasing to 250 divisions reduces the ratio to approximately 10. The final native MAPDL preflight completes with zero warnings and zero errors.

2.2 Why a 0.10-degree sector

Two angular divisions retain three-dimensional topology and explicit convection faces, while 240 radial rows provide the resolution needed for progressive removal. The sector cannot represent large-scale circumferential nonuniformity. It is appropriate for a local axisymmetric response; any crack, joint, or localized jet requires a wider angular domain.

3 Physics and thermochemical model

3.1 Energy conservation in the solids

The model solves transient heat conduction in the phenolic and aluminium. In the reactive phenolic, the local equation is

$$\rho(\alpha)c_p(T, \alpha) \frac{\partial T}{\partial t} = \nabla \cdot [k(T, \alpha) \nabla T] - \rho \Delta H_p \dot{\alpha} - \rho \dot{q}_{g,s}, \quad (1)$$

where $\Delta H_p > 0$ is the endothermic pyrolysis enthalpy and $\dot{q}_{g,s}$ is sensible cooling by released gas. In aluminium the last two terms vanish and the properties are $k = 167 \text{ W m}^{-1} \text{ K}^{-1}$, $c_p = 896 \text{ J kg}^{-1} \text{ K}^{-1}$, and $\rho = 2700 \text{ kg m}^{-3}$. The conformal interface enforces continuity of temperature and normal heat flux; no thermal contact resistance is added.

3.2 Pyrolysis kinetics and evolving properties

The state variable $0 \leq \alpha \leq 1$ is zero in virgin material and one in the char residue. The single reaction follows

$$\dot{\alpha} = A \exp\left(-\frac{E_a}{RT}\right) (1 - \alpha)^n, \quad A = 333 \text{ s}^{-1}, \quad E_a = 64\,081 \text{ J mol}^{-1}, \quad n = 1. \quad (2)$$

`UserMatTh` integrates this law analytically at the increment midpoint temperature:

$$\alpha_{j+1} = 1 - (1 - \alpha_j) \exp[-k_r(T_{j+1/2})\Delta t]. \quad (3)$$

It requests a MAPDL cutback when $\Delta\alpha > 0.05$. Density and thermal properties are interpolated between virgin and char states,

$$X(T, \alpha) = (1 - \alpha)X_v(T) + \alpha X_c(T), \quad X \in \{\rho, k, c_p\}. \quad (4)$$

The reference densities are $\rho_v = 1250 \text{ kg m}^{-3}$ and $\rho_c = 600 \text{ kg m}^{-3}$, while the provisional pyrolysis enthalpy is $\Delta H_p = 418 \text{ kJ kg}^{-1}$. The k and c_p tables vary with temperature and must ultimately be replaced or validated with measurements of the manufactured material.

3.3 Gas generation and reduced evacuation model

Local condensed-mass loss generates pyrolysis gas:

$$\dot{\omega}_g''' = Y_g(\rho_v - \rho_c)\dot{\alpha}, \quad Y_g = 1 \text{ in v0}. \quad (5)$$

Gas mass is integrated over all live elements and retained in the balance. The model solves neither porosity, pore pressure, nor Darcy flow; it assumes that gas reaches the hot surface rapidly. Its sensible enthalpy is nevertheless removed from the solid through

$$\dot{q}_{g,s} = f_g \frac{\dot{\omega}_g'''}{\rho} c_{p,g} \max(T - T_{g,\text{ref}}, 0), \quad (6)$$

where $f_g = 1$, $c_{p,g} = 1600 \text{ J kg}^{-1} \text{ K}^{-1}$, and $T_{g,\text{ref}} = 300 \text{ K}$.

The total gas flow reduces convective transfer through a blowing correction:

$$B_h = \frac{\dot{m}_g'' c_{p,g}}{h_0}, \quad h_{\text{eff}} = h_0 \frac{B_h}{\exp(B_h) - 1}, \quad q_{\text{conv}}'' = h_{\text{eff}}(T_g - T_s). \quad (7)$$

The limit $B_h \rightarrow 0$ recovers $h_{\text{eff}} = h_0$. This closure retains the protective effect of outgassing without claiming to resolve pore flow or the chamber boundary layer.

3.4 Boundary conditions and chamber pressure

The hot face receives $T_g = 2230.85^\circ \text{C}$ and $h_0 = 1000 \text{ W m}^{-2} \text{ K}^{-1}$. The framework includes

$$q_{\text{rad}}'' = \varepsilon_h \sigma (T_{g,\text{rad}}^4 - T_s^4), \quad (8)$$

but $\varepsilon_h = 0$ in v0 so that an uncharacterized hot-side radiation load is not introduced. The outer face exchanges with a 22°C ambient through convection $h = 8 \text{ W m}^{-2} \text{ K}^{-1}$ and radiation $\varepsilon = 0.25$. Axial and lateral sector faces are adiabatic, representing a uniform local periodic slice.

The specified nominal internal pressure is $p_c = 800 \pm 200 \text{ psi}$, or $5.52 \text{ MPa} \pm 1.38 \text{ MPa}$, with a 600–1000 psi (4.14–6.89 MPa) envelope. The nominal value is explicit in the configuration; the two bounds are sensitivity cases to be run once the other inputs are frozen.

3.5 Char consumption and recession

Pyrolysis creates char but does not automatically remove material. A second law acts only on the hot face. Its provisional kinetic rate is placed in series with a mass-transfer limit:

$$\dot{m}_{\text{kin}}'' = \dot{m}_{\text{ref}}'' \exp\left[-\frac{E_c}{R} \left(\frac{1}{T_s} - \frac{1}{T_{\text{ref}}}\right)\right] \left(a_r \frac{p_c}{p_{\text{ref}}}\right)^m, \quad (9)$$

$$\dot{m}_{\text{tr}}'' = \dot{m}_{\text{cap}}'' a_r \frac{p_c}{p_{\text{ref}}}, \quad \dot{m}_c'' = \left(\frac{1}{\dot{m}_{\text{kin}}''} + \frac{1}{\dot{m}_{\text{tr}}''}\right)^{-1}. \quad (10)$$

The exploratory values are $E_c = 100 \text{ kJ mol}^{-1}$, $T_{\text{ref}} = 1500 \text{ K}$, $m = 0.5$, $\dot{m}_{\text{ref}}'' = 0.01 \text{ kg m}^{-2} \text{ s}^{-1}$, and $\dot{m}_{\text{cap}}'' = 0.05 \text{ kg m}^{-2} \text{ s}^{-1}$. The applied rate is also bounded by the amount of char actually formed, by 20 % of char mass per macro-step, and by available net power:

$$\dot{m}_{c,\text{applied}}'' \leq \frac{\max(q_{\text{conv}}'' + q_{\text{rad}}'' - q_{\text{cond}}'', 0)}{H_{\text{eff}}}, \quad H_{\text{eff}} = 35 \text{ MJ kg}^{-1}. \quad (11)$$

Recession is therefore driven by disappearance of solid mass, not by $\alpha = 1$ alone.

3.6 Row removal and numerical coupling

Every macro-step restores the last accepted state, computes loads on the currently exposed face, solves the MAPDL thermal problem, exports converged fields, checks balances, and only then accepts a checkpoint. A row is scheduled for EKILL only when its residual solid mass falls below 0.1 % of its initial virgin mass. The row is deactivated at the next macro-step and hot-side conditions move to the newly exposed face. Residual mass and energy are logged before the topology changes.

The nominal step is 0.0125 s, reducible to 0.0005 s; the reference horizon is 7 s. After each converged state the controller closes

$$m_0 = m_{\text{solid}} + m_g + m_c + m_{\text{ejected residue}} \quad (12)$$

and audits energy using native element terms and the internal sinks imposed by `UserMatTh`. In MAPDL 2026 R1, `SOLID278/NMISC,39` remains zero for these user sources; the controller therefore reconstructs the pyrolysis sink independently from $\Delta\alpha$ and the gas sensible sink from state variable 5, while retaining the native channel as a diagnostic. A mass error above $10^{-6}m_0$, an energy error above 5 %, or a hot-load discrepancy above 2 % blocks the next solve. Convection is integrated from the documented face heat rates `NMISC,29` (hot face 5) and `NMISC,17` (outer face 3). The aggregate `NMISC,40` rate, which under-reports the load here with `UserMatTh`, is retained as a secondary diagnostic only.

Validity domain. Chamber pressure is now known, but gas composition, reactive fractions, and char-oxidation kinetics are not. The consumption law therefore remains `EXPLORATORY_NOT_CALIBRATED`. The calculation represents 3D conduction, pyrolysis, evolving properties, enthalpy sinks, blowing, surface consumption, and discrete element removal. It does not represent the chamber flow, resolved porous transport, mechanical erosion, cracking, delamination, thermal expansion, or structural stress.

4 Primary quantities and balances

4.1 Remaining thickness

After $N_k(t)$ rows have been accepted for removal,

$$s_{\text{removed}}(t) = N_k(t) \Delta r_{\text{ph}}, \quad s_{\text{remaining}}(t) = s_0 - s_{\text{removed}}(t), \quad s_0 = 4.7625 \text{ mm}.$$

The discrete thickness resolution is therefore 19.843 75 μm . The hot-wall coordinate is $R_h(t) = R_i + s_{\text{removed}}(t)$.

4.2 Consumed phenolic mass

Three quantities must not be conflated:

1. cumulative gas mass produced by pyrolysis;
2. carbon mass consumed at the exposed surface;
3. virgin-equivalent mass of geometrically removed rows.

For a full ring of length L , the last quantity is

$$m_{\text{removed},0}(t) = \rho_v \pi L \left[(R_i + s_{\text{removed}}(t))^2 - R_i^2 \right].$$

The history retains both sector values and values multiplied by 3,600. Every step closes mass between initial mass, remaining solid, pyrolysis gas, consumed carbon, and ejected residual. An error beyond tolerance automatically prevents the next solve.

5 Interim results as of September 24, 2026

5.1 Accepted state at 5.2500 seconds

This documentation snapshot ends at accepted step 405; the planned horizon remains 7 s. No future solver result is included. Only one row has been removed: 0.019 843 75 mm removed and 4.742 656 25 mm still present geometrically. The remaining thickness is not necessarily virgin material. The mean hot-face temperature reaches 1505.14 °C; this is not the maximum temperature of the model.

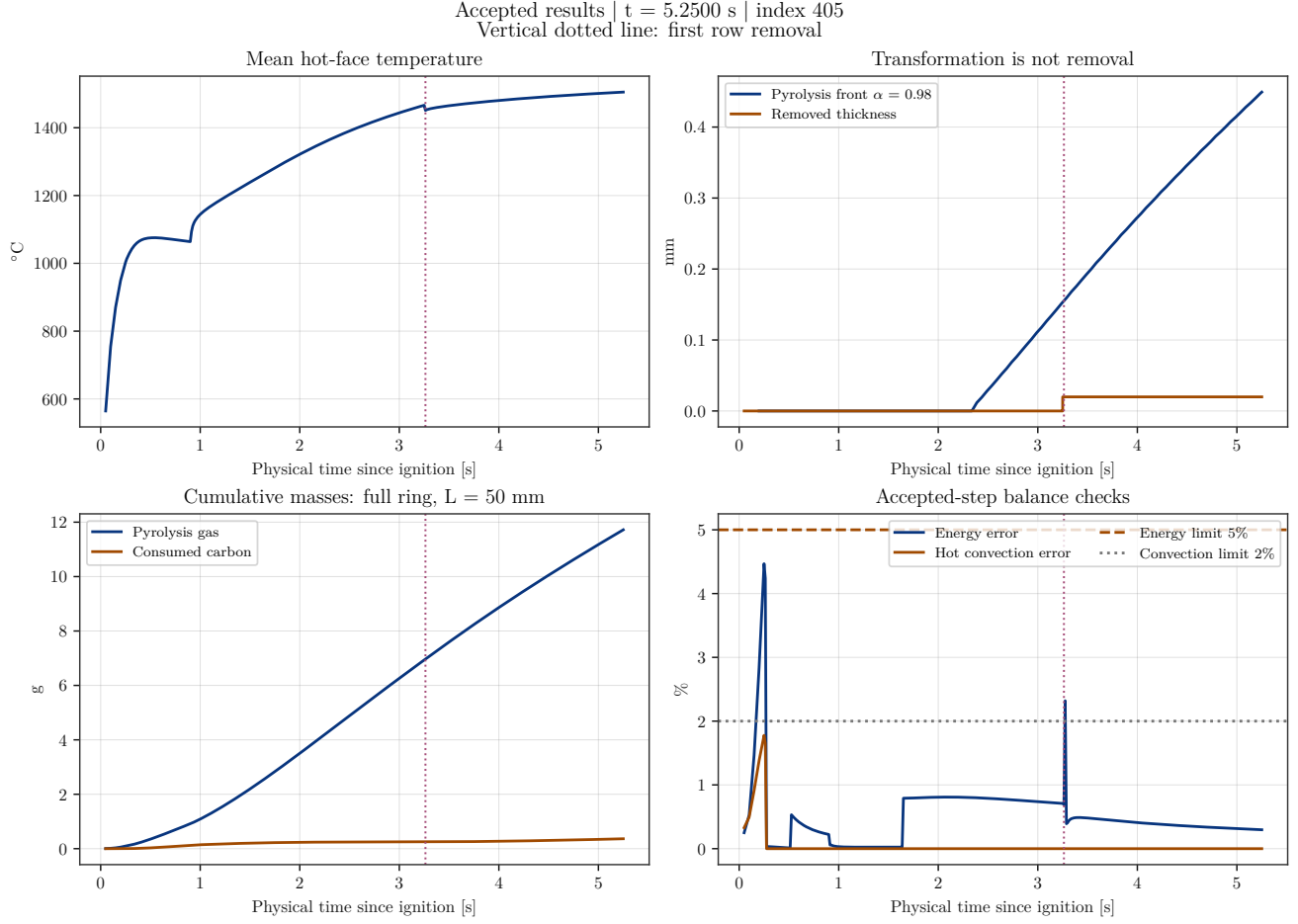


Figure 3. Accepted history: mean hot-face temperature, pyrolysed depth and geometric removal, cumulative masses, and balance errors. Masses are scaled to a full ring of 50 mm length. The vertical dotted line marks the first row removal.

At the latest state, the relative energy error is 0.2968% (5% limit) and the relative hot-convection error is 6.01×10^{-9} (0.02 limit). Cumulative gas mass is 11.7197 g and consumed-carbon mass is 0.364 54 g, on the full-ring scale. All accepted audits are below both limits. This validates these numerical checks, not the exploratory material laws.

5.2 Pyrolysis-front progression

For consistency with the previous estimates, the front is defined by $\bar{\alpha} \geq 0.98$, where $\bar{\alpha}$ is the volume-weighted mean of the 500 elements in a radial row. We retain the contiguous zone from the original bore and interpolate linearly between row centres bracketing the threshold. Already removed rows retain their last accepted conversion in this original-bore depth measure.

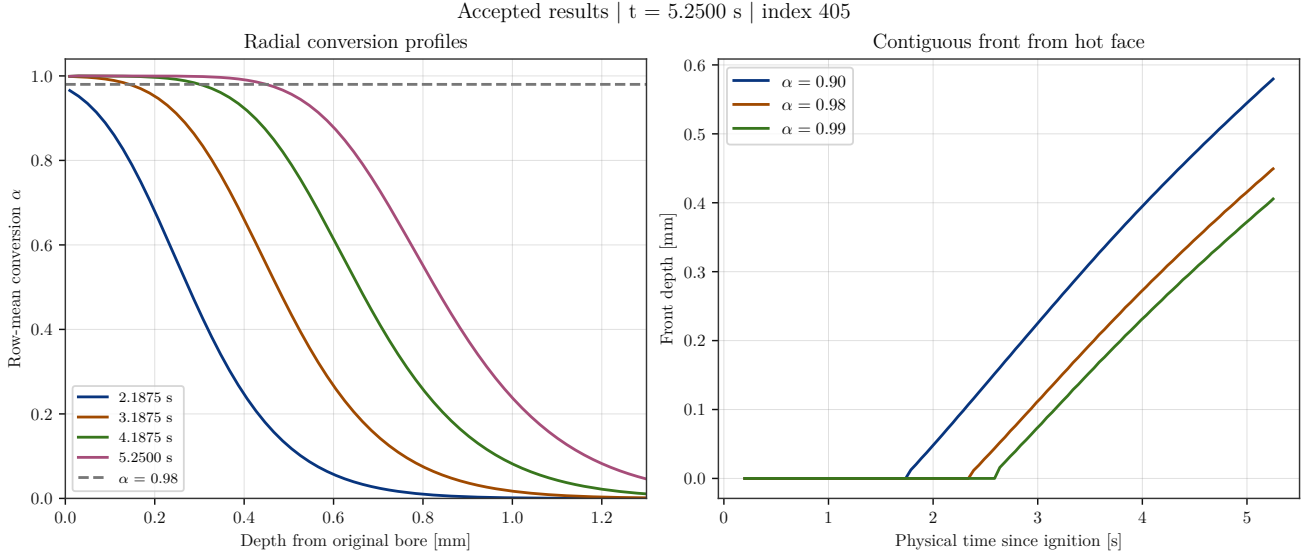


Figure 4. Conversion profiles and depths of the 90, 98, and 99 % fronts. Profiles show the first 1.3 mm of the original 4.7625 mm thickness. Front histories sample every fourth checkpoint, supplemented by the latest accepted state and the previous comparison checkpoint.

At 5.2500 s, the 98 % front is at 0.449 28 mm, or 9.43 % of the initial thickness. The 90 and 99 % fronts are at 0.579 58 mm and 0.405 56 mm, respectively. The threshold is a post-processing definition: it changes neither the material law nor the removal rule. **Pyrolysed therefore means neither burned nor geometrically removed.**

5.3 25, 50, and 75 percent projections

A least-squares straight line is fitted to the 98 % front depth over the latest available 0.6625 s, using the same sampling. The current fit uses 15 points from 4.5875 s to 5.2500 s. Its fitted speed is $0.13647 \text{ mm s}^{-1}$, versus $0.14557 \text{ mm s}^{-1}$ at the previous 4.6625 s assessment.

Table 2. Physical times since ignition, not wall-clock calculation durations.

Pyrolysed initial thickness	Previous [s]	Updated [s]	Change
25 %	10.30	10.68	+3.7 %
50 %	18.48	19.40	+5.0 %
75 %	26.66	28.13	+5.5 %

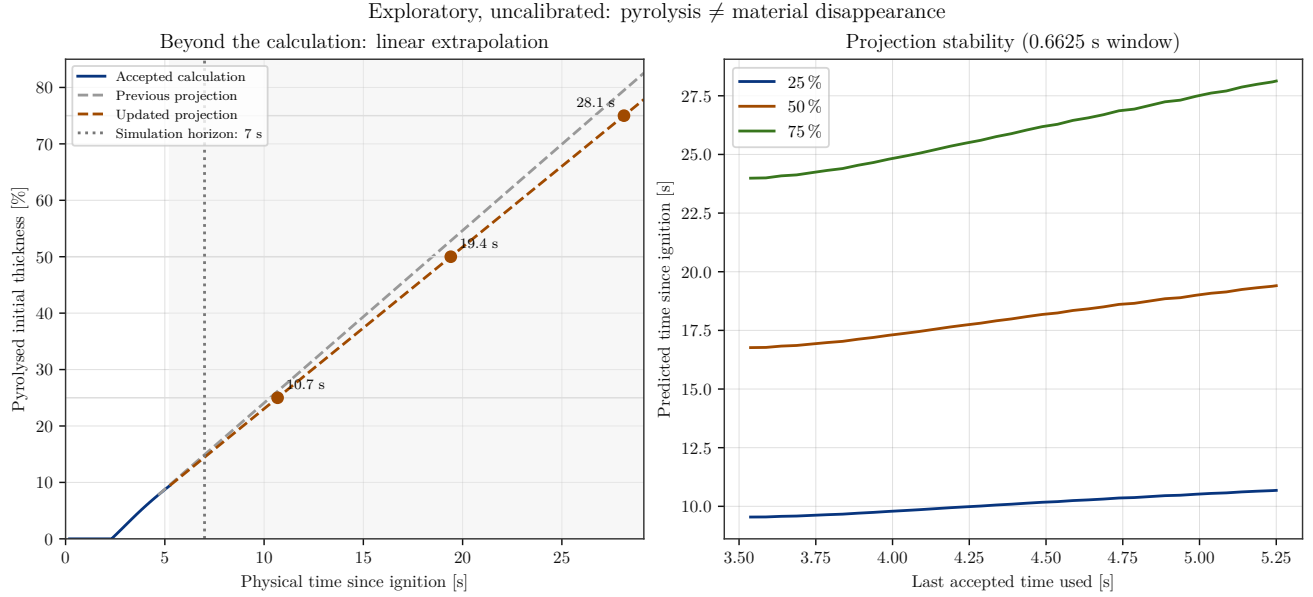


Figure 5. Left: accepted calculation in solid blue and extrapolations in dashes; shading extends beyond available results. Right: drift in predicted crossing times as new checkpoints enter the fitting window.

The projection at 7 s is approximately 14.46 % pyrolysed. The 25/50/75 % estimates remain of the same order, but rise as the front slows: they have not stabilised in the sense of predictive convergence. All these crossings lie beyond the currently authorised horizon. These are uncalibrated linear extrapolations, with no physical confidence interval; they do not demonstrate motor survivability. Uncertainty grows substantially at 50 and 75 %.

5.4 Sources and reproducibility

The `newmotor_v0_history.csv` history is a view of immutable JSON audits for each accepted step: time, thicknesses, masses, surface temperature, conversion, loads, and balances. Figures read accepted audits and conversion checkpoints directly, never a step still being exported. The reduced snapshot and source SHA-256 hashes are stored in `docs/analysis/accepted_results_snapshot.json`. The `make_results_figures.py` script rebuilds figures of-fline; its `-capture` option reads the runtime without changing the calculation. Text and tables in this edition are frozen at the same index 405.

6 Windows execution, pause, and resume

All large files are isolated in Git-ignored `v0/ansystmp`. The `v0` directory is mapped to drive `V:` in Windows, and the runtime is `V:\textbackslash ansystmp\textbackslash windows`. It has no reference to the older project. The `UserMatTh` source and DLL are copied locally.

Before the first calculation, `launch_v0.ps1 -Preflight` verifies input hashes, mesh, the local `1055@localhost` license, and reads the complete deck without `SOLVE`. The September 20, 2026 preflight confirms 229,665 nodes, 152,000 elements, zero warnings, and zero errors. The transient run was subsequently explicitly launched on 10 ranks. On September 22, step 247 at 3.275 s was recovered after a false `POST1` export rejection following `EKILL`, without relaxing thresholds. The controller requires all live identifiers, rejects duplicates and foreign identifiers, and ignores only surplus rows already declared dead. The failure archive and `RECOVERY_PROVENANCE.json` preserve provenance.

Launch requires explicit `-Run` and uses 10 DMP ranks. The script refuses to start when `ANSYS/MPI` already exists. Six restart generations are retained. `request_pause.ps1` requests a stop at the next accepted macro-step; `-Run -Resume` is allowed only from a `PAUSED_AT_CHECKPOINT` state whose inputs remain unchanged.

7 Run status and next decisions

Item	Status
Two-volume geometry	created and archived in a native MAPDL database
Structured mesh	validated: 229,665 nodes, 152,000 hexahedra
Windows preflight	complete, 0 warnings, 0 errors, no solve
Solver inputs	sealed with SHA-256
Chamber pressure	800 $\pm$ 200 psi, nominal 5.52 MPa
Pause/resume	prepared at accepted checkpoints
French/English reports	compiled from the same parameters
Transient calculation	running; frozen state at 5.2500 s, index 405

To qualify the results, the actual burn duration, gas composition, and measured phenolic properties are still required. Without them, `v0` is deliberately a seven-second exploratory reference. Its structure is nonetheless ready to answer the primary question: how much phenolic was transformed or removed, and how much sound thickness remains at each time?